

Problem Set 1 — Solutions

Problem 1: Spaces and Norms

Part 1

2. Newtonian forces acting on an object

- Newtonian forces can be represented as belonging to $\mathbb{R}^2$ for 2D objects or $\mathbb{R}^3$ for 3D objects.
- A value carrying both magnitude and direction, drawn from a reference point — e.g. a force of $[3, 4]$ on an object in the 2D plane, drawn from the origin.

4. Angle of a robot's elbow joint Represented by $\mathbb{R} \bmod 360$ is *not* a vector space. It fails on the scalar-distribution axiom, e.g. take $a = \frac{1}{2}$, $u = 200$, $v = 300$:

$$\begin{aligned}a(u + v) &= \frac{1}{2} \cdot (500 \bmod 360) = \frac{1}{2} \cdot 140 = 70, \\au + av &= \frac{1}{2} \cdot 200 + \frac{1}{2} \cdot 300 = 100 + 150 = 250.\end{aligned}$$

So $a(u + v) \neq au + av$, and the scalar-distribution axiom fails.

Diversion: a vector is a function

The vector $[1, 2, 5, 3, 2, 6]$ can be interpreted as a function: just define $f(k) = v_k$. The index k is the input and the output (the value of the function) is v_k .

This is exactly the idea that tells apart the three cases above:

- A finite vector $[v_1, \dots, v_n]$ is a function on a *finite* domain $k \in \{1, \dots, n\}$.
- An infinite sequence $(v_1, v_2, v_3, \dots)$ is a function on the *countably infinite* domain $k \in \mathbb{N}$.
- A continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ (like $\sin x$) has an *uncountably infinite* domain.

So *every* vector space we have met is really a space of functions — the vectors differ only in how large (and how “dense”) their index set is.